Cubic Regularization When Both Derivatives Are Noisy

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Stochastic Cubic Regularization for Fast Nonconvex Optimization

Key: tripuraneni2018stochastic Tags: S / E; B Tier: Must read

Survey: Escape methods

Spreadsheet priority: None; independent recommendation: Must read

The paper in one minute

Question: Can cubic regularization certify approximate second-order stationarity using only stochastic gradients and stochastic Hessian-vector products?

Model/Method: Build sampled cubic models, solve them approximately with perturbed gradient descent, and refine the final model more accurately when its predicted decrease is small.

Finding: Theorem1 and Corollary1 give leading total oracle complexity $\tilde{O}(\varepsilon^{-7/2})$ under smoothness, concentration and subsolver conditions.

Why we care: The method controls how derivative errors affect model progress and stopping. It does not require the exact objective to be evaluated for each acceptance decision.

1. Why an indefinite quadratic needs a stabilizer

At x, a second-order Taylor model uses a linear gradient term and a quadratic Hessian term. If the Hessian has a negative eigenvalue, the quadratic model can decrease without bound along that direction. Adding a positive norm-cubed term makes large moves expensive. This is a regularized second-order Taylor model, not a claim that $||s||^3$ is the true third-order Taylor derivative.

$$m(s) = g^T s + (1/2)s^T Bs + (\rho/6)||s||^3.$$

Here g estimates $\nabla f(x)$, and B estimates $\nabla^2 f(x)$. Access to B is through products Bv, computed from a fixed Hessian minibatch for the current model. No dense Hessian is required. The model gradient is $g + Bs + (\rho/2)||s||s$. The cubic coefficient convention matters: it is $\rho/6$ here, whereas Carmon-Duchi's subproblem paper uses $\rho/3$.

2. Conditions and the target

The expected f is lower bounded, has an ℓ-Lipschitz gradient and a ρ-Lipschitz Hessian. Assumption2 controls stochastic gradient and Hessian errors by variance bounds σ_1^2,σ_2^2 and almost-sure bounds M_1,M_2. Independent minibatches estimate the two derivatives. The target is $||\nabla f|| \le \varepsilon$ and $\lambda\min(\nabla^2 f) \ge -\sqrt{(\rho\varepsilon)}$. Despite the source's approximate-local-minimum terminology, these inequalities alone do not exclude every degenerate saddle.

Two approximation errors. Sampling approximates the true derivatives. The inner solver then only approximately minimizes that sampled model. A theorem must control both; a small inner gradient alone is not enough to justify the complete outer algorithm.

3. Worked example: what noisy curvature does to a cubic step

Pedagogical illustration, constructed for these notes.

Use $f(s) = \cos s$ at $x = 0$. Then $f'(0) = 0$ and $f''(0) = -1$, while global gradient and Hessian Lipschitz constants can both be chosen as one. The exact centered cubic with $\rho = 1$ is $q(s) = -s^2/2 + |s|^3/6$. Its nonzero minimizers are $s = \pm 2$, each with $q = -2/3$.

Now suppose independent derivative estimates give $g = 0.01$ and $B = -0.9$. The sampled model is $m(s) = 0.01s - 0.45s^2 + |s|^3/6$. These values can arise from bounded unbiased component perturbations $F(s;\zeta) = (1+\zeta_h)\cos s + \zeta_g \sin s$, with symmetric $\zeta_h = \pm 0.1$ and $\zeta_g = \pm 0.01$.

The positive g selects the negative direction. Put $s = -r$, $r \geq 0$. Setting the derivative of $-0.01r - 0.45r^2 + r^3/6$ to zero gives $r^2/2 - 0.9r - 0.01 = 0$.

$$r = 0.9 + \sqrt{0.83} \approx 1.811043; \; m(-r) \approx -0.504055.$$

The exact cubic at this point is $q(-r) \approx -0.649939$, while the actual loss change is $\cos(r) - 1 \approx -1.237943$. Thus this particular sampled step makes real progress; a random sampled model is not automatically an upper bound on the true loss.

The model-error bound is $|m(s) - q(s)| \leq |g|r + |B+1|r^2/2 = 0.01r + 0.05r^2 \approx 0.182104$. Its exact difference here is about 0.145883, safely below the bound.

Near termination the characteristic step length is $\sqrt{(\varepsilon/\rho)}$. Gradient error of order ε times this length and Hessian error of order $\sqrt{(\rho\varepsilon)}$ times its squared length are both order $\varepsilon^{3/2}/\sqrt{\rho}$. That is the same scale as useful cubic decrease, so the hidden accuracy constants must be small. Fixed arbitrary derivative errors cannot support arbitrarily precise certification.

4. The outer method uses predicted progress

At each outer iteration, Algorithm1 draws fresh gradient and Hessian minibatches, fixes the sampled model, and asks Cubic-Subsolver for a step and its model change Δm. Large predicted decreases imply true objective decreases on the derivative-concentration event and under Condition1. When Δm is too small, Cubic-Finalsolver refines the model before returning a point. Simply returning the last inexact step can fail the desired stationarity test.

Condition1 allows either sufficiently large model and true-function decrease, or a controlled step length plus sufficiently accurate model minimization in the relevant regime. The condition is an analytical contract; it does not require the implementation to query the unknown expected objective. The proposed subsolver supplies the contract using gradients of the sampled cubic.

5. Why the total cost is $\varepsilon^{-7/2}$

Lemma2 in the supplement bounds the sampled gradient error by $c_1\varepsilon$ and the Hessian operator error by $c_2\sqrt{(\rho\varepsilon)}$. Matrix concentration is uniform over v, so the same Hessian batch can be reused at adaptively chosen inner directions. In the small-ε variance-dominated regime, the leading counts are:

Work per scale	Leading dependence
Outer model iterations	$\tilde{O}(\sqrt{\rho}\,\Delta f\,\varepsilon^{-3/2})$.
Gradient minibatch n_1	$\tilde{O}(\sigma_1^2\,\varepsilon^{-2})$.
Hessian minibatch n_2	$\tilde{O}(\sigma_2^2/(\rho\varepsilon))$.
Inner products per model $T(\varepsilon)$	$\tilde{O}(\ell/\sqrt{(\rho\varepsilon)})$.

Each gradient batch is computed once per model; each Bv query costs n_2 component products. Thus total work is outer count times $[n_1 + n_2T(\varepsilon)]$. With fixed parameters, the gradient term scales as $\varepsilon^{-7/2}$, and the main inner-product term as ε^{-3}. The exact theorem also keeps M_1/ε and $M_2/\sqrt{(\rho\varepsilon)}$ terms when the variance-dominated simplification does not apply. Final refinement is accounted for and does not change the leading total exponent.

For a requested whole-run failure δ, AppendixB.5 divides the probability budget among concentration and subsolver events over all outer iterations. A per-batch failure probability cannot simply be declared to be the whole-run failure probability.

6. Solving the sampled model

Algorithm2 takes an analytic Cauchy step along -g when the gradient is large. Otherwise it perturbs g slightly and runs gradient descent on the cubic. The perturbation addresses a hard case: at g = 0, a zero-initialized deterministic inner gradient method stays at zero even when B has negative curvature. This is randomness in the inner model, distinct from data-sampling noise. Small predicted decrease triggers Algorithm3's more accurate final solve.

7. What the experiments support

The synthetic problem combines a one-dimensional W-shaped piecewise cubic with $10x_2^2$ and has saddle Hessian eigenvalues -0.2 and 20 at zero. Figure2 compares progress by oracle calls. The MNIST experiment uses a softplus autoencoder with encoder 784→512→256→128→32 and symmetric decoder, final sigmoid and reconstruction squared loss. Figure3 compares cubic, SGD, AdaGrad and SVRG+Oja.

SupplementD reports ten inner iterations in experiments, gradient batch100 and Hessian batch10 for the autoencoder, and a 55,000/5,000 training-validation split. Hyperparameters are selected under a two-million-oracle-call budget; the full batch-size grid was not searched for every method. The curves show favorable tested performance. Plateaus alone do not independently certify that every displayed point is a saddle, and oracle-call comparisons are not equal wall-clock comparisons.

8. Limits and survey connections

The guarantees require calibrated derivative accuracy and a suitable inner solver. The finite experimental inner count is not a blanket certificate that Condition1 holds on every model. The output is approximate second-order stationarity, not a universal global minimum. Unlike Kohler2017's acceptance test using exact objective values, this fully stochastic construction reasons from model decrease. Compared with Fang2019, its leading exponent uses a stronger curvature oracle under different noise assumptions.

9. Quick reference

Item	Meaning
ρ/6	Cubic coefficient; model gradient uses ρ/2.
$n_1 + n_2T$	One gradient batch, repeated sampled Hessian products.
Finalsolver	Needed to certify the returned inexact point.

Source: main PDF1-9, Theorem1/Corollary1 and Algorithms1-3; all publisher supplement1-15 (printed11-25), including proof and experimental protocols. Numerical cosine example constructed for teaching.

Papers Master - Our notes (copy): Fully stochastic cubic regularization attains approximate second-order stationarity with leading total Õ(ε^-7/2) gradient/Hessian-vector calls. Requires ε-scale gradient accuracy, √(ρε)-scale Hessian accuracy, an appropriate inner solver and final refinement. Model progress replaces exact objective acceptance; experiments do not certify universal global optimality.

Reading recommendation: Read the derivative-accuracy scales, meta-algorithm and final refinement. This is the fully stochastic bridge from cubic models to finite-time second-order stationarity.